Reliability of Early Pancreatic Cancer Biomarkers under Heterogeneous Controls and Limited Data

Spectrum Sensitivity Analysis and a LYVE1/REG1B Illustration

ChatGPT

Abstract

Background. Retrospective pancreatic cancer biomarker studies contain few stage-I cases and controls whose clinical composition may differ from that of the intended testing population. We assess the reliability of the resulting diagnostic claims.
Methods. We derive sharp false-positive risk bounds under uncertain healthy/benign composition and bounded within-group shift, and a simultaneous binomial calibration rule. Four synthetic scenarios were simulated with 700 replications each. Archived LYVE1 and REG1B results were assessed using reported aggregate counts.
Results. Under correctly specified shift assumptions, false-positive cap violations occurred in 78.4%, 2.9%, and 0% of simulations using pooled, spectrum-adjusted, and simultaneous-confidence calibration, respectively; mean stage-I sensitivities were 26.3%, 13.6%, and 2.9%. Unmodeled benign-marker displacement caused 42.9% violations for the simultaneous-confidence rule. In the retrospective archive, conservative LYVE1 and REG1B thresholds detected 2/16 and 1/16 stage-I cases, respectively.
Conclusions. Control composition, assumptions about transport, and limited early-stage samples materially affect biomarker evaluation. These analyses quantify uncertainty but do not validate population screening performance.

1 Introduction

Pancreatic ductal adenocarcinoma (PDAC) is difficult to detect before clinical diagnosis. Biomarker signals may be weak in early disease, while benign pancreatic and hepatobiliary conditions can produce overlapping measurements. Consequently, an assay that distinguishes diagnosed cancer from healthy volunteers may perform differently among patients undergoing clinical assessment. The intended-use population and its noncancer spectrum determine which specificity estimates are relevant [1, 2].

Debernardi and colleagues evaluated urinary LYVE1, REG1B, and related markers in a retrospective study of 590 specimens [3]. A later investigation examined prediagnostic urine samples [4]. Neither sample design directly identifies the false-positive rate of a threshold in a representative screening population. Reporting guidance emphasizes prespecified clinical use, informative controls, and external validation [1, 5]. Population-wide PDAC screening of asymptomatic average-risk adults is not currently recommended [6].

We examine the reliability of biomarker evaluation rather than propose a new clinical test. Specifically, we quantify how uncertain healthy/benign control composition and bounded within-group differences affect false-positive risk. We then account for sampling uncertainty when choosing among candidate marker thresholds and distinguish what archived aggregate data can establish from what would require individual-level or representative target data. The approach builds on established diagnostic-inference and exact-binomial methods [2, 7, 8].

2 Estimands and clinical assumptions

For an individual in a prespecified target setting, let $D = 0$ denote reference-standard-confirmed absence of PDAC, and let $D = 1$ denote PDAC at the time of biomarker collection. Let Y_m be a numerical score for marker $m \in \{\text{LYVE1}, \text{REG1B}\}$. At threshold t, a positive test is $\delta_{m,t} = \mathbf{1}\{Y_m \geq t\}$. The target noncancer false-positive rate is $\text{FPR}_T(m,t) = \mathbb{P}_T(Y_m \geq t \mid D = 0)$; target stage-I sensitivity is $\text{Se}_{I,T}(m,t) = \mathbb{P}_T(Y_m \geq t \mid D = 1, \text{stage I})$.

The source noncancer population is divided into healthy (H) and benign-disease (B) specimens. The corresponding threshold-positive probabilities are $p_{H,m}(t)$ and $p_{B,m}(t)$. The target benign share among noncancer patients is w, which is not the benign share in the case–control sample and is not assumed identifiable from it. Our clinical sensitivity analysis specifies an externally chosen interval $w \in [\underline{w}, \overline{w}]$, rather than estimating a fictitious screening fraction from study recruitment.

Two distinct shifts must be distinguished. A composition shift changes w but leaves biomarker distributions within groups unchanged. A within-group shift additionally changes a score law, for instance because benign conditions, assay settings, or comorbidity patterns differ across studies. A sensitivity parameter $\Gamma_g \geq 1$ bounds the Radon–Nikodym density ratio of a target subgroup distribution with respect to the corresponding source distribution. Without independent target data, neither w nor Γ_g is learned here; displayed ranges and guarantees are explicitly conditional on these assumptions.

Stage-I and stage-II patients are not exchangeable for the purpose of estimating early detection. We report the two stages separately. A stage-weighted sensitivity could be evaluated under hypothetical clinical stage proportions, but case counts sampled by design do not estimate those proportions. We also do not assume that stage-conditional case distributions automatically transport from retrospective detection to prediagnostic screening.

3 Statistical methods

3.1 Control-spectrum risk bounds

If control distributions within both strata transport unchanged, the target risk is

$$F_w(m,t) = (1-w)p_{H,m}(t) + w p_{B,m}(t), \qquad w \in [\underline{w}, \overline{w}]. \tag{1}$$

Thus a marker that seems accurate against healthy controls can exhibit worse performance in a population containing more benign disease. This is not a label-prevalence adjustment;

both components in (1) are noncancer participants.

Proposition 3.1 (Exact composition envelope)**.** *For fixed subgroup laws and prespecified* $[\underline{w}, \overline{w}]$*, the sharp target-control interval at* (m, t) *is*

$$[\underline{F}, \overline{F}] = \Big[\min_{v \in \{\underline{w}, \overline{w}\}} \{(1-v)p_H + vp_B\},\ \max_{v \in \{\underline{w}, \overline{w}\}} \{(1-v)p_H + vp_B\} \Big]. \tag{2}$$

Both endpoints and every intermediate value are attainable by allowed target compositions.

Proof. Equation (1) is affine in w and attains its minimum and maximum at the endpoints of a compact interval. Continuity establishes intermediate attainability, and each w corresponds to a genuine mixture of the two fixed control laws. □

Within-group differences. Write P_g for the source score distribution and Q_g for its target counterpart. Consider the sensitivity model

$$\Gamma_g^{-1} \le \frac{dQ_g}{dP_g} \le \Gamma_g \quad P_g\text{-a.s.}, \qquad \Gamma_g \ge 1. \tag{3}$$

The assumption excludes new target subgroup support outside source support; its clinical plausibility must be investigated rather than inferred from the mathematical guarantee. Let

$$\phi_\Gamma(p) = \min\{\Gamma p,\ 1 - (1-p)/\Gamma\}. \tag{4}$$

Theorem 3.2 (Sharp pointwise clinical-spectrum envelope)**.** *If both control groups satisfy* (3)*, then the maximal target false-positive rate of a fixed* (m, t) *is exactly*

$$U_m(t) = \max_{v \in \{\underline{w}, \overline{w}\}} \Big[(1-v)\phi_{\Gamma_H}\{p_{H,m}(t)\} + v\phi_{\Gamma_B}\{p_{B,m}(t)\} \Big]. \tag{5}$$

If target scores may also differ from an intermediate density-reweighted source score by at most d_g *almost surely, the same formula with* $p_{g,m}(t - d_g)$ *remains an attainable upper bound when arbitrary measurable couplings and the deterministic* $+d_g$ *displacement are admitted. Sharpness is pointwise in* t*: a single shifted distribution need not attain the maxima at all thresholds simultaneously.*

Proof. For an event A, (3) implies $Q_g(A) \le \Gamma_g P_g(A)$ and $Q_g(A) = 1 - Q_g(A^c) \le 1 - P_g(A^c)/\Gamma_g$. Hence $Q_g(A) \le \phi_{\Gamma_g}\{P_g(A)\}$. To attain this value, put $Q_g(A) = \phi_{\Gamma_g}(p)$ and $Q_g(A^c) = 1 - \phi_{\Gamma_g}(p)$, rescaling P_g by constants on A and A^c; both rescaling constants lie in $[1/\Gamma_g, \Gamma_g]$. Cases $p = 0$ or 1 follow trivially. This establishes exact eventwise bounds independently in both groups. Maximizing their weighted sum over w uses Proposition 3.1. With displacement $|Y_T - Y'| \le d_g$, $\{Y_T \ge t\} \subseteq \{Y' \ge t - d_g\}$. Equality for the upper risk is obtainable with the deterministic allowable transformation $Y_T = Y' + d_g$, completing the argument. □

The parameters Γ_B and d_B represent different uncertainties. The former can change the relative frequency of existing source values; the latter permits a bounded numerical movement of scores. One cannot justify either by noting that a mathematical optimization problem has a solution. Clinical estimates in our archived tables used assumptions stronger than pure composition uncertainty and should not be reinterpreted as unconditional guarantees.

3.2 Finite-sample calibration and incomplete archives

Suppose two markers are measured in separate or jointly observed groups of n_H and n_B independent noncancer participants and a prespecified grid $\mathcal{T}$ of K finite thresholds is available. Dependence between markers measured on the same people is allowed. Let $N_{g,m}(t)$ count control scores $\geq t$, and let $\overline{p}_{g,m}(t)$ be its one-sided Clopper–Pearson upper confidence limit with noncoverage budget

$$\delta = \alpha/(2MK), \qquad M = 2 \text{ markers}, \tag{6}$$

where 2 counts the control strata. For $N = k < n$,

$$\overline{p}(k, n, \delta) = \text{Beta}^{-1}(1 - \delta; k + 1, n - k), \qquad \overline{p}(n, n, \delta) = 1. \tag{7}$$

The distinction between marker multiplicity and subgroup multiplicity is explicit: there are $2MK$ binomial events in the family, without an unjustified independence assumption between thresholds.

Theorem 3.3 (Simultaneous control of target false-positive risk)**.** *Assume iid source observations within each labeled control group, the given $[\underline{w}, \overline{w}]$ and Γ_g limits, and the same score definitions and thresholds in target use. For each (m, t), substitute $\overline{p}_{g,m}(t)$ into (5) to obtain $\widehat{U}_m(t)$. Then*

$$\mathbb{P}\{\text{FPR}_T(m, t) \leq \widehat{U}_m(t) \text{ for every } m, t \in \mathcal{T}\} \geq 1 - \alpha. \tag{8}$$

Consequently any data-dependent selection $\widehat{(m, t)}$ from the set $\{(m, t) : \widehat{U}_m(t) \leq \beta\}$ has target $\text{FPR}_T \leq \beta$ with probability at least $1 - \alpha$, even if marker choice uses the same observed stage-I cases. The always-negative rule is an exact fallback when this set is empty. The result is conditional on correct shift and composition restrictions, and does not certify target sensitivity.

Proof. For a fixed subgroup, marker, and threshold, the exceedance count is binomial marginally, so the one-sided Clopper–Pearson limit covers the source exceedance probability with probability at least $1 - \delta$. A union bound over the $2MK$ events gives a joint event with probability at least $1 - \alpha$. On this event, $p_{g,m}(t) \leq \overline{p}_{g,m}(t)$ for all indices. Since ϕ_Γ and the endpoint weighted combinations are nondecreasing in each argument, Theorem 3.2 yields $\text{FPR}_T(m, t) \leq \widehat{U}_m(t)$ for every rule simultaneously. Selection after observing the sample cannot invalidate a simultaneous event. A constant-zero decision rule has exactly zero FPR regardless of the assumptions. □

This confidence construction deliberately prioritizes interpretability and finite-sample validity over power. At the clinical control sample sizes, it may select high thresholds that retain few stage-I cancers. Because rules are chosen using stage-I observations, their in-sample stage-I sensitivity cannot be taken as an unbiased future sensitivity estimate. One can also allocate a held-out case cohort for sensitivity estimation, but the present data do not support a precise, prospective stage-I performance claim.

Archived aggregate counts. An archived report may retain only the control denominators n_H, n_B and the pooled number k of positive tests, rather than counts by subgroup. Let k_H range over all compatible integers:

$$\max(0, k - n_B) \le k_H \le \min(k, n_H), \qquad k_B = k - k_H. \tag{9}$$

Even for empirical reweighting there is missing information. Define

$$\widehat{F}(w, k_H) = (1 - w)\frac{k_H}{n_H} + w\frac{k - k_H}{n_B}. \tag{10}$$

Proposition 3.4 (Sharp aggregate-data reweighting range)**.** *For the archive just described, the full range of achievable empirical reweighted false-positive fractions has endpoints equal to the minimum and maximum of* (10) *over the four pairs* (w, k_H) *obtained by setting* w *and* k_H *to their respective extreme values in* (9) *and* $[\underline{w}, \overline{w}]$*. The interval contains no finite-sample statistical coverage statement and does not bound an unobserved target population's true risk.*

Proof. For each fixed w, (10) is affine in k_H, so its extrema over integer k_H occur at the smallest or largest feasible integer. At either such endpoint, the function is affine in w, so its extrema occur at $\underline{w}$ or $\overline{w}$. Each corner is an actual allocation of k recorded positives among the $n_H + n_B$ controls and an allowed mixture share, establishing attainability of both extrema. No statement is made about unattainable intermediate continuous values when count allocations are discrete. □

Stage-specific sensitivity and predictive value. Stage-I and stage-II detection should be reported separately. If a target case mixture with stage-I share η were credibly specified, with stage-conditional sensitivities transporting unchanged, then $\mathrm{Se}_T(\eta, t) = \eta\,\mathrm{Se}_I(t) + (1 - \eta)\,\mathrm{Se}_{II}(t)$; endpoint optimization is identical to Proposition 3.1. Neither the target stage proportion nor such transport is identified by the retrospective counts. Finally, the positive predictive value for a target prevalence π is

$$\mathrm{PPV}(\pi, t) = \frac{\pi\,\mathrm{Se}_T(t)}{\pi\,\mathrm{Se}_T(t) + (1 - \pi)\,\mathrm{FPR}_T(t)}, \tag{11}$$

when the denominator is positive. Prevalence cannot be estimated from the deliberate case–control sampling proportions. Calculations from (11) must therefore be expressly scenario-based, not reported as measured clinical PPV.

4 Monte Carlo study

The Monte Carlo study is intended to expose clinical limitations of a modest method, not to imitate the empirical distribution of LYVE1 or REG1B. Two synthetic, normalized biomarker scores are constructed. The LYVE1-like source score has normal laws $N(-0.55, 0.85^2)$ for healthy controls, $N(0.05, 1.05^2)$ for benign controls, and $N(0.70, 0.95^2)$ for stage-I cases. The REG1B-like score has $N(-0.35, 0.95^2)$, $N(0.15, 1.00^2)$ and $N(0.58, 0.95^2)$, respectively. These are synthetic parameters, not estimates from Debernardi et al.

For each marker, the benign target distribution is generated by reweighting its source density by a factor $r_{\mathrm{hi}} = 1.30$ above score 0.55 and a normalizing factor $r_{\mathrm{lo}} = (1 - r_{\mathrm{hi}} p_{\mathrm{hi}})/(1 - p_{\mathrm{hi}})$ below it, where p_{hi} is the source probability above 0.55. Thus both factors fall in $[1/1.5, 1.5]$ and the correctly specified model uses $\Gamma_H = 1$, $\Gamma_B = 1.5$, $w \in [0.2, 0.8]$, with generating $w = 0.7$. The false-positive target is $\beta = 0.10$. This is a methodological test cap, not a recommendation for PDAC screening, which may require far higher specificity [1].

For each draw we choose among 19 prespecified thresholds from -0.30 to 3.30 for both markers, selecting the feasible rule with the largest apparent stage-I sensitivity. We compare: (i) a naive empirical pooled-control rule; (ii) a group-composition and within-benign-shift plug-in envelope; and (iii) the simultaneous-confidence envelope of Theorem 3.3 with $\alpha = 0.05$. True target FPR and true stage-I sensitivity are evaluated from the generating distributions, not from the training specimens. An always-negative fallback is allowed. The standard configuration has $n_H = 183, n_B = 208, n_I = 16$; an enlarged-control configuration uses $n_H = n_B = 900, n_I = 50$.

Two deliberate failures of maintained assumptions are considered. In the first, the target benign share is $w = 0.88$ and the high-score reweighting is 1.90. In the second, the benign target score has an additive shift of $+1.0$, outside the assumed bounded density-ratio model. All scenarios are hypothetical. The reproducible Python run uses seed 240918 and 700 replications per setting. Replication-level records and generating code accompany the manuscript.

Table 1: Synthetic Monte Carlo results (700 replications per scenario). Violations indicate a true target false-positive rate above 10%; sensitivity refers to the generating stage-I distribution.

Scenario	Calibration	Violation (%)	FPR (%)	Stage-I Se (%)
Clinical-size, covered	Pooled	78.4	12.17	26.28
	Spectrum plug-in	2.9	5.66	13.62
	Simultaneous	0.0	1.04	2.86
Wrong spectrum/ratio	Pooled	97.3	21.82	26.73
	Spectrum plug-in	46.3	10.15	13.73
	Simultaneous	0.1	2.05	3.14
Unmodeled displacement	Pooled	100.0	37.69	26.57
	Spectrum plug-in	95.7	23.54	13.67
	Simultaneous	42.9	8.03	3.25
Expanded controls, covered	Pooled	99.7	14.45	31.32
	Spectrum plug-in	0.0	6.47	15.78
	Simultaneous	0.0	3.90	10.07

No violation among 700 trials is an observation, not proof that the true violation probability is zero. The guarantee in Theorem 3.3 is 95% under correct assumptions; an overly conservative empirical violation frequency is compatible with that guarantee. Conversely, under a wrong model the theorem is inapplicable, and the unmodeled-displacement setting directly exhibits failures. The source sample composition is also systematically different from target composition, explaining why pooled calibration can deteriorate rather than

improve when the sample size grows. The principal substantive message is the cost of honest extrapolation when false-positive limits matter.

5 LYVE1 and REG1B: retrospective illustration

Debernardi et al. reported 590 retrospective urine samples: 183 healthy controls, 208 benign-disease controls and 199 PDAC cases, including 16 stage-I and 86 stage-II cases [3]. The complete original data are available in the publication's supporting material. In the present execution environment, however, the patient-level workbook was not materialized. We therefore reanalyzed previously generated Phase-IV clinical CSV outputs rather than pretending to have re-run patient-level marker extraction. The accompanying Python package reconstructs the aggregate table without imputing individual participants.

The archived Phase-IV methods were a pooled plug-in threshold and a conservative split-exact hybrid threshold under a previously specified benign shift model ($\Gamma_B = 2$ and a 0.1 displacement in log-transformed score units). These thresholds were selected in the retrospective study workflow, with no independent clinical validation cohort. We have not reestimated them here. Table 2 reports observed classifications and a new aggregate-data reweighting range computed from the archived counts by Proposition 3.4. The latter deliberately discards unknown subgroup-positive counts, reflecting the information actually available to this specific reanalysis. It does not claim those counts are absent from the publicly released patient-level dataset.

Table 2: Retrospective LYVE1 and REG1B classifications from archived analyses. The final column is a hypothetical empirical reweighting range for benign-control shares of 20%–80%, not a confidence interval.

Marker	Archived rule	Cutoff	Stage I	Stage II	Control +	Reweight (%)
LYVE1	Pooled	5.36	3/16	45/86	35/391	3.37–15.30
LYVE1	Conservative	8.97	2/16	11/86	3/391	0.29–1.31
REG1B	Pooled	140.17	6/16	42/86	39/391	3.75–17.05
REG1B	Conservative	517.01	1/16	8/86	4/391	0.38–1.75

The conservative thresholds detect only $2/16 = 12.5\%$ and $1/16 = 6.25\%$ of the observed stage-I cases, respectively. Even treating these thresholds as fixed independently of case observations, two-sided exact binomial 95% intervals would be approximately 1.6%–38.3% and 0.16%–30.2%. These are reference binomial intervals and must not be interpreted as post-selection clinical confidence intervals without proof that the observed stage-I sample was independent of the rule-selection step. The marked contrast between stage-I and stage-II observed detection also argues against reporting a single early-stage summary without stratification.

The last column of Table 2 is not an interval for the true screening FPR. For instance, three pooled positive LYVE1 controls could be distributed as $(k_H, k_B) = (3, 0)$ or $(0, 3)$, and the two extreme choices of benign share assign different weights to those observed positives. This gives 0.29%–1.31% from the same three observed outcomes, before introducing any

statistical uncertainty or within-group shift. The width is a diagnostic of insufficiently detailed reporting, not a clinical estimate of future transport.

The study confirms neither that a urinary marker can screen average-risk adults nor that the target false-positive rate is certified at the archived threshold. It shows an interpretable trade-off: lowering the apparent control-positive fraction by choosing a more stringent threshold may substantially reduce observed stage-I case detection. A clinically defensible analysis must distinguish its control population, avoid using source case fractions as target prevalence, report stage-I uncertainty, and declare how marker selection was handled. These requirements are consistent with the PDAC biomarker-development and STARD literature [1, 5].

6 Discussion

The analysis identifies three separate limitations of retrospective biomarker evaluation. First, healthy and benign-disease controls need not represent the same clinical spectrum as noncancer patients in the target setting. Second, even a specified target composition cannot resolve unmeasured changes within a control stratum. Third, the small number of stage-I cases makes sensitivity estimates imprecise. Our risk envelopes separate the first two assumptions, while simultaneous calibration addresses sampling uncertainty in source controls.

The simulations illustrate a trade-off between false-positive control and sensitivity. Under correctly specified shift restrictions, simultaneous calibration avoided observed cap violations but detected few synthetic stage-I cases. The spectrum-adjusted plug-in rule retained more sensitivity with less reliable calibration. The displaced-benign scenario shows that sampling guarantees cannot correct incorrect transport assumptions. These results describe the specified synthetic models, not the clinical performance of the urine markers.

The archived LYVE1/REG1B analysis makes the reporting problem concrete. More conservative thresholds reduced observed pooled control positives while also reducing stage-I detections. When subgroup-positive counts are unavailable, alternative allocations of the same pooled positives yield different empirical risks after changing the assumed benign-control share. The resulting ranges do not represent population confidence intervals. Access to the original subgroup counts would remove this archival uncertainty, but not the need for external validation.

The method has limitations. Two noncancer strata cannot capture the full diversity of benign diagnoses, and density-ratio bounds exclude new target support. Composition and shift parameters are sensitivity assumptions rather than quantities learned from these data. The simultaneous guarantee applies to a prespecified finite collection of rules and addresses false-positive risk, not target sensitivity. Both the simulation and the clinical illustration lack a representative unlabeled target cohort; the latter also uses archived aggregate results rather than independently processed patient-level records. A subsequent validation study would need clinically relevant controls, prespecified thresholds, assay comparability, and independent stage-I or prediagnostic outcome assessment [1, 4, 5].

AI-generation declaration

This manuscript, mathematical arguments, code, and reported analyses are fully AI-generated and may contain errors. Executing computational checks does not constitute independent verification of the proofs, clinical interpretation, references, or originality.

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